

BAIT3153 SOFTWARE PROJECT MANAGEMENT

City Light Rail Train (LRT) operates from 6.00am mornings until 12.00am midnight daily and serves 20 stations of a dedicated route in Kuala Lumpur. For the past one month, a customer service helpdesk received the passengers' complaint about the train's arrival time was delayed at the station during peak hours. Mr. Eric, as an operation manager of LRT has decided to develop an e-schedule mobile application that allows the passengers to check the schedules information and receive the notifications. The e-schedule application to be developed consists of schedule and notification modules. A budget of RM 200,000 has been approved for the project with a duration of 10 weeks. The system specification of e-schedule application focuses on the core features of the modules and high user involvement is encouraged to collect user requirements. The formation of the project team consists of ten junior staff from the Information Technology department who will be working on this project.

Answer the following questions based on the given scenario.

Question 1

- a) Table 1 shows the progress of project e-schedule with a total allocated budget of RM 200,000. Assume that the project is now in the 7th week and as of today, you have paid out RM 183,000 to the contractors. Contractors are those who performed the tasks shown in Table 1. Using an earned value management technique, calculate the *Planned Value (PV)*, *Earned Value (EV)*, *Schedule Variance (SV)* and *Cost Variance (CV)* for this project. Based on the results of your calculations, give a detailed analysis on the overall performance of this project. (6 + 5 marks)

Table 1: Task progress of project e-schedule

Task ID	Estimated Duration (in weeks)	Estimated Cost (RM)	Status (percent completed)
A	1	15,000	100%
B	1	15,000	100%
C	1	30,000	100%
D	2	50,000	100%
E	2	60,000	100%
F	2	20,000	50%
G	1	10,000	0%

Note: show the working in your calculation. SV formula is $EV - PV$, CV formula is $EV - AC$.

Task ID	Estimated Duration (weeks)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	15,000	100%	$15,000 * 100\% = 15,000$
B	1	15,000	100%	$15,000 * 100\% = 15,000$
C	1	30,000	100%	$30,000 * 100\% = 30,000$
D	2	50,000	100%	$50,000 * 100\% = 50,000$
E	2	60,000	100%	$60,000 * 100\% = 60,000$
F	2	20,000	50%	$20,000 * 50\% = 10,000$
G	1	10,000	0%	$10,000 * 0\% = 0$
Total	10	200,000	-	$15,000 + 15,000 + 30,000 + 50,000 + 60,000 + 10,000 + 0 = \text{RM}180,000$

PV

Activity A+B+C+D+E = 7 weeks

= 15,000 + 15,000 + 30,000 + 50,000 + 60,000

= RM 170,000

AC = RM 183,000

EV = 15,000 + 15,000 + 30,000 + 50,000 + 60,000 + 10,000 + 0

= RM180,000

SV = EV - PV

= 180,000 - 170,000

= RM 10,000 (positive ahead of schedule)

CV = EV - AC

= 180,000 - 183,000

= - RM 3,000 (over budget)

SV is positive (RM10,000) means that the project is ahead of schedule. In the 7th week, the actual value of work completed is more than the value of work planned to be completed.

CV is negative (-RM3000), which means that the project is over budget. In the 7th week, the actual value of work completed is less than the amount paid.

- b) Table 2 shows the function point measurement of the e-schedule project:

Table 2: Function point measurement

Measurement parameter	Weighting factor	Count
Number of inputs	Simple (1)	2
Number of outputs	Simple (2)	2
Number of inquiries	Average (3)	3
Number of files	Average (4)	4
Number of external interfaces	Complex (7)	5

Note: FP formula is $\text{Count Total} \times [0.65 + 0.01 \sum Fi]$, an adjustment value $\sum Fi$ is 35.

- (i) Calculate the Function Point (FP) for the above project. Show your workings in your calculation. (8 marks)

Measurement parameter	Weighting factor	Count	C * WF
Number of inputs	Simple(1)	2	1 * 2 = 2
Number of outputs	Simple(2)	2	2 * 2 = 4
Number of inquiries	Average(3)	3	3 * 3 = 9
Number of files	Average(4)	4	4 * 4 = 16
Number of external interfaces	Complex(7)	5	7 * 5 = 35
			Count Total = 66

The Function Point for the about project, FP

$$= \text{Count Total} \times [0.65 + 0.01 \sum Fi]$$

$$= 66 * [0.65 + 0.01 (35)]$$

$$= 66 * [1]$$

$$= 66$$

- (ii) 'Function oriented metric is suitable to assess the project scopes'. Do you agree with this statement? Provide a justification to support your answer. (6 marks)

[Total: 25 marks]

Yes, I agree. Metrics, such as Function Points (FP), measure the functionality delivered to the user based on system inputs, outputs, user interactions, files, and external interfaces. They help quantify the functional size of a software project, providing a clear understanding of its scope. Since these metrics don't depend on the programming language and tools, they can be applied early for effort and cost estimation, making them valuable for project planning and scope control. By focusing on user-visible features, they ensure the scope reflects actual business requirements. Thus, function-oriented metrics offer a simple and reliable way to understand and manage the project scope.

BAIT3153 SOFTWARE PROJECT MANAGEMENT

Question 2

- a) Eric has identified a project objective as '*increase passenger's satisfaction by reducing waiting time during peak hours*'. A well-defined objective should have specific, measureable, achievable, relevant, and time-constrained (SMART) criteria.

Consider any **TWO (2)** SMART criteria that fulfill the above objective. Justify your suggested criteria with an explanation. (8 marks)

Specific & Measurable

Criterion: Reduce average passenger waiting time during peak hours by 30% within 6 months.

Justification: This criterion is specific which is reducing waiting time and measurable since 30% reduction can be tracked through data collection such as ticketing system logs or passenger feedback.

- b) Which software process model is most suitable for the e-schedule mobile application: *Spiral* or *Rapid Application Development* model? Provide **TWO (2)** justifications that can be applied for the suggested software process model used in this project. (1 + 6 marks)

Rapid Application Development (RAD)

1. Faster Development – Since the system needed to be completed within 70 days which is 10 weeks, RAD allows for quick prototyping and iterative releases, enabling stakeholders (passengers, operators) to test features like scheduling and notifications early. This ensures the system meets actual user needs.
2. Lack Of Experts – Since we have 10 junior IT staff, RAD is more suitable than the Spiral model. Spiral model required experts in the development teams and required long development time which is not suitable for the 10 weeks time constraints.

- c) Gantt charts are used to illustrate the sequence of activities in a project. Figure 1 is an example of a Gantt chart.
 Assume that Task ID A will start in week 1 of January 2025.
 Redraw the Gantt chart in Figure 1 using the predecessor information provided in Table 3 to illustrate the interdependencies between project activities. Identify the month and week for Task ID G will be completed.

Figure 1: An example of a Gantt chart.

	January 2025				February 2025				March 2025	
Task ID	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
A										
B										
C										
D										
E										
F										
G										

Table 3: Project activity and its predecessor

Task ID	Task Name	Duration (in weeks)	Predecessor
A	Meeting with stakeholders	1	None
B	Requirement gathering	1	A
C	Business modelling	1	A
D	Data modelling for schedule and notification modules	2	B, C
E	Code for schedule and notification modules	2	D
F	System testing	2	E
G	Conduct user acceptance test	1	F

(8 + 2 marks)

[Total: 25 marks]

	January 2025				February 2025				March 2025	
Task ID	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
A										
B										
C										
D										
E										
F										
G										

Task G completes in March 2025, Week 9.

BAIT3153 SOFTWARE PROJECT MANAGEMENT**Question 3**

- a) Eric has identified some potential risks of the e-schedule project in the Table 4.

Table 4: Project risk, probability and cost

No.	Risk	Probability of occurrence (0 to 1)	Cost	Risk Exposure
1	Tight schedule	0.3	2000	
2	Unclear system specification	0.4	3000	
3	Poor quality software caused by junior members	0.5	4000	
4	Conflict occurs in team	0.2	1000	

Note: Risk Exposure (RE) metric is *Probability X Cost*

- (i) Calculate the *Risk Exposure* (RE) for each risk in Table 4. Rank the risks according to their priority (i.e. from highest to lowest RE). Show the working in your calculation. (6 marks)

No.	Risk	Probability of occurrence (0 to 1)	Cost	Risk Exposure
1	Tight schedule	0.3	2000	$0.3 * 2000 = 600$
2	Unclear system specification	0.4	3000	$0.4 * 3000 = 1200$
3	Poor quality software caused by junior members	0.5	4000	$0.5 * 4000 = 2000$
4	Conflict occurs in team	0.2	1000	$0.2 * 1000 = 200$

Ranking by RE (highest to lowest):

1. Risk 3 (2000)
2. Risk 2 (1200)
3. Risk 1 (600)
4. Risk 4 (200)

(ii) Based on the results of your calculation in Question 3 a) (i), identify **ONE (1)** risk with the highest risk exposure. Propose a suitable risk *mitigation, monitoring and management actions* for the identified risk. Provide an explanation in your answer. (1 + 6 marks)

Poor quality software caused by junior members.

1. Mitigation actions (prevent / reduce probability & impact)
 - Pair juniors with seniors for real-time code review, enforce standards through pull requests requiring senior approval, and allocate senior developers to critical modules. These steps reduce poor-quality code and lower costs by catching defects early.
2. Monitoring actions (detect & measure)
 - Quality can be monitored through metrics such as defect density, escaped defects and code review stats with rework trends. These ongoing measures provide early warnings, enabling corrective action before release.
3. Management (contingency & response)
 - Contingency measures include reserving budget and schedule buffers for rework or extra QA, bringing in external experts for critical modules, and reprioritizing scope to delay non-essential features. A customer communication plan ensures stakeholders are informed of impacts and revised timelines. These controls limit business impact and provide practical recovery options if mitigation fails.

b) Change management is to ensure that system evolution is properly managed in the change process. Evaluate **TWO (2)** factors in change analysis that influence the change request from City LRT stakeholders. Provide an explanation in your answers. (6 marks)

1. The benefits of the change
 - The proposed change is analyzed in terms of the value it delivers to stakeholders, such as improved passenger satisfaction, safety, or revenue generation. If the benefits are significant and directly support the LRT's service objectives, the change request is more likely to be approved.
 - For example, a change request to add a real-time train arrival feature in the app benefits passengers by reducing waiting anxiety. Since the benefits strongly improve service quality, this change has a high chance of approval.
2. The costs of making the change
 - Every change involves costs such as development effort, resource allocation, training, maintenance, and potential delays to the project schedule. The feasibility of the change depends on whether the benefits outweigh these costs. A change with high cost but low benefit is unlikely to be approved.
 - For example, a change request to redesign the whole app interface near the end of the project would be very costly. The stakeholders may reject the request due to high cost compared to limited gain.

- c) The e-schedule project implementation is considered to be *which type of critical system*?
Provide a justification with an example in your answer. (6 marks)

[Total: 25 marks]

- Mission-critical system.
- The system directly supports daily transit operations (scheduling, notifications) and passenger flow. Failure or prolonged outage during peak hours will disrupt operations, cause major delays, revenue loss, and serious passenger dissatisfaction. It threatens the mission of providing reliable transit service.
- For example, If the e-schedule app fails during the morning rush hour, passengers might not know the correct train times, leading to long waits and crowded platforms. This would disrupt the smooth running of the transit service.

5

BAIT3153 SOFTWARE PROJECT MANAGEMENT

Question 4

- a) Performing quality management activities will incur costs in the e-schedule mobile application. Given below are two categories of quality costs:
- Appraisal costs
 - Failure cost

Using the quality management activities given below, categorised each of them into one of the categories listed above. Explain which category is likely to be *most costly* to the City LRT Company. Justify your answer.

Complaint resolution, conduct technical reviews, poor reputation, perform system testing and provide help line support inquiries.

(5 + 4 marks)

1. Appraisal costs (prevention, detection, evaluation before release):
 - Conduct technical reviews
 - Perform system testing
2. Failure costs (costs due to defects after release):
 - Complaint resolution
 - Poor reputation
 - Provide help line support inquiries

Failure costs are likely the most costly because defects after release affect real passengers, leading to high complaint handling, customer support, and damage to company reputation. Reputational loss can reduce customer trust and ridership which has long-term financial impact.

b) Given below are three dimensions of system dependability:

- Availability
- Reliability
- Security

Select **TWO (2)** dimensions listed above that are likely to be *most important* to an e-schedule mobile application of City LRT. Briefly describe the selected dimensions and explain their inter-dependencies. (8 marks)

Dimensions:

1. Availability – It ensures the mobile app is accessible whenever passengers need to check schedules. Downtime directly disrupts travel planning which increases passenger dissatisfaction.
2. Reliability – It ensures the app provides correct, consistent, and accurate schedule information. Incorrect data such as wrong train times can cause missed trains and loss of passenger trust.

Interdependencies:

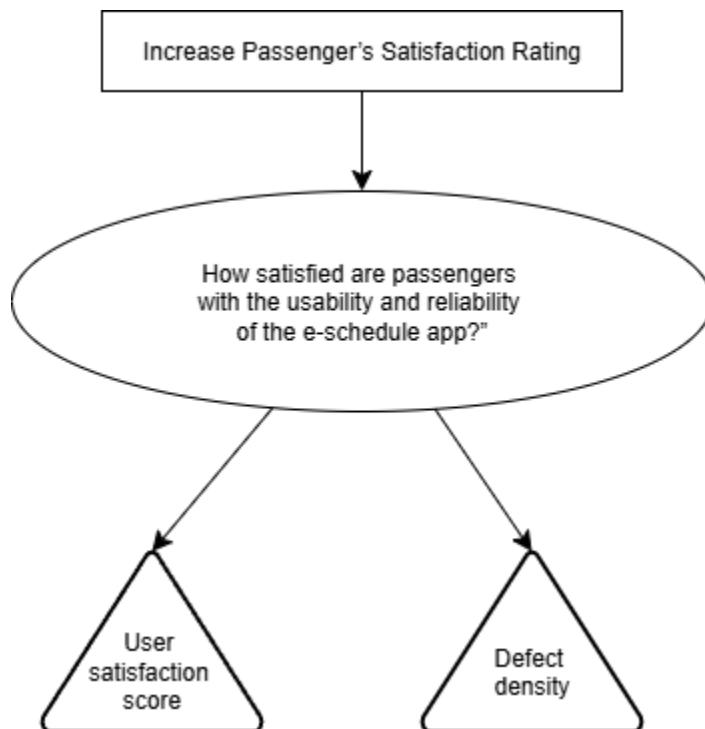
Availability without reliability is useless. For example, an app that runs but shows wrong info harms passengers. Reliability without availability is also ineffective. For example, accurate info is pointless if the system is down. Both must work together to support dependable passenger services.

- c) Eric has identified the project goal as '*Increase Passenger's Satisfaction Rating*' to support the software process improvement initiative.

Create a Goal Question Metric (GQM) proposal for the given goal and include the following information:

- Formulate **ONE (1)** question to assess the given goal.
- Propose **TWO (2)** relevant software metrics and explain how each of the software metrics can be applied to the suggested question. (8 marks)

[Total: 25 marks]



1. User Satisfaction Score is the average rating from surveys or in-app feedback that passengers give on a scale such as 1–5. This metric directly measures perceived satisfaction with usability, design, and usefulness of the app. By tracking scores, management can assess whether changes improve passenger satisfaction and identify areas needing improvement.
2. Defect Density is the number of confirmed defects or failures per thousand lines of code (KLOC). This metric reflects the technical quality of the software. Tracking defect density over time shows if quality practices are reducing errors. Fewer errors make the app more reliable, which helps keep or improve passenger satisfaction.